

Criterion A: Knowing and Understanding

Question 1 – **State** which of the following correctly describes the particle arrangement and movement in a solid.

- A. Particles are packed closely together and vibrate in fixed positions.
- B. Particles are loosely arranged and can flow past one another.
- C. Particles have no movement at all and are perfectly still.
- D. Particles are spread far apart and move freely in all directions.

Question 2 – **Identify** the temperature at which pure water changes from a liquid to a solid (freezes).

- A. -100°C
- B. 0°C
- C. 37°C
- D. 100°C

Question 3 – **Identify** the change of state in which a liquid changes into a gas from its surface at temperatures below its boiling point.

- A. Condensation
- B. Melting
- C. Sublimation
- D. Evaporation

Question 4 – **Identify** the process by which a solid changes directly into a gas without passing through the liquid state.

- A. Boiling
- B. Condensation
- C. Sublimation
- D. Freezing

Question 5 – **Identify** the state of matter that has both a fixed shape and a fixed volume.

- A. Gas
- B. Plasma
- C. Liquid
- D. Solid

Question 6 – Outline the three states of matter. For each state, state whether it has a fixed shape and a fixed volume.

State of matter	Fixed shape?	Fixed volume?
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Solid

Liquid

Gas

Question 7 – Identify the correct name for each change of state and complete the table.

Starting state	Final state	Name of change
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Solid	Liquid	_____
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Gas	Liquid	_____
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Liquid	Solid	_____
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Question 8 – Describe what happens to the particles in ice when it is heated and melts into liquid water. Refer to particle energy, movement and arrangement in your answer.

Question 9 – Water droplets form on the outside of a cold glass of juice on a humid day.

Identify the statement that best explains this observation.

- A. The juice inside the glass evaporates and forms droplets on the outer surface.
- B. Water from inside the glass seeps through the glass walls to the outside.
- C. The glass absorbs moisture from the air and stores it inside the glass.
- D. Water vapour in the air cools against the cold glass and turns into liquid water droplets.

Question 10 – Engineers leave small gaps between sections when building a metal bridge.

Identify the statement that best explains why these gaps are necessary.

- A. Gaps reduce the weight of the bridge, making it safer for heavy vehicles.
- B. Metals expand when heated, so gaps allow the bridge sections to expand in hot weather without buckling or cracking.
- C. Gaps allow rainwater to drain away so that the bridge does not become slippery.
- D. Metals contract when heated, so gaps prevent the sections from pulling apart in hot weather.

Question 11 – After swimming, Mia notices that she feels cold when she gets out of the pool, even on a warm day.

Explain why Mia feels cold using your knowledge of evaporation and particle energy.

Question 12 – A bottle of perfume is opened at one end of a classroom. After a short time, students at the other end of the classroom can smell it.

Explain why the perfume smell spreads across the room. Refer to the kinetic theory of particles and diffusion in your answer.

Question 13 – Salt is spread on icy roads during winter.

Using your knowledge of melting points, **identify** the statement that best explains why salt helps to keep roads clear of ice.

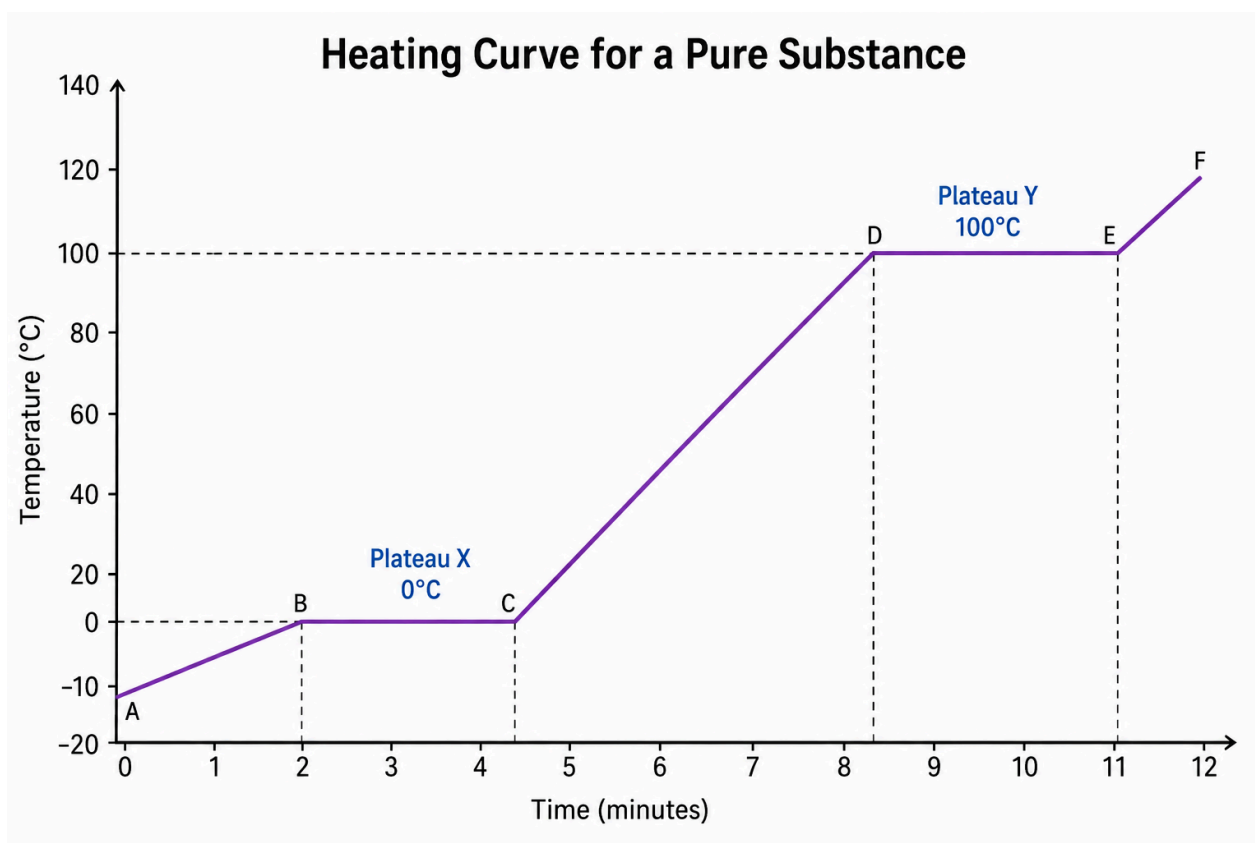
- A. Salt absorbs heat from sunlight and directly heats the road above 0°C , melting the ice.
- B. Salt lowers the melting/freezing point of water, allowing ice to melt at temperatures below 0°C .

- C. Salt raises the melting point of ice, causing the ice to remain solid for longer.
D. Salt reacts chemically with ice to produce a new substance that is not slippery.

Question 14 –The graph below shows how the temperature of a sample of a pure substance changes as it is heated over time.

The graph shows:

- Temperature rises from -10°C to 0°C from 0–2 minutes.
- A flat plateau X occurs at 0°C from 2–4.5 minutes.
- Temperature rises from 0°C to 100°C from 4.5–8.5 minutes.
- A flat plateau Y occurs at 100°C from 8.5–11 minutes.
- Temperature rises above 100°C after 11 minutes.

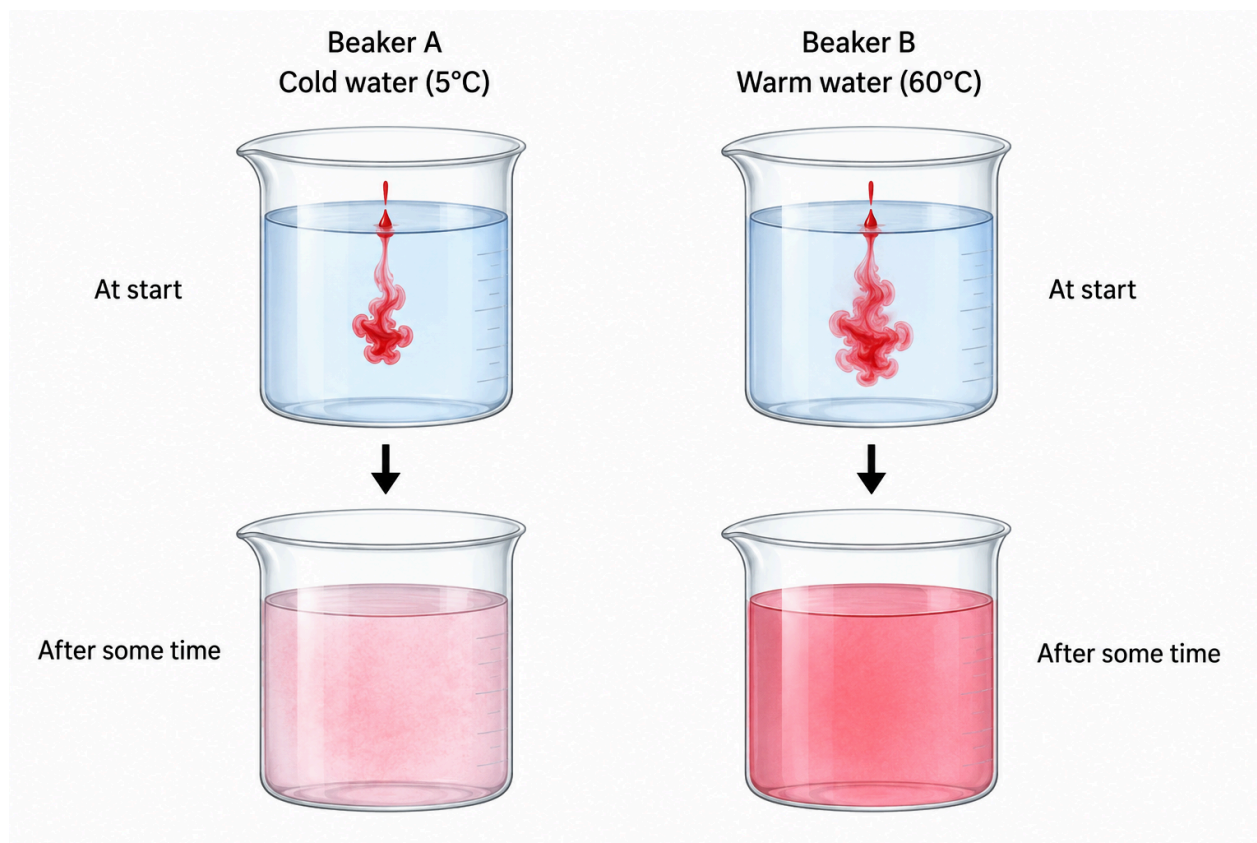


- (a) Identify the change of state occurring at plateau X.
- (b) Identify the change of state occurring at plateau Y.
- (c) Explain why the temperature does not increase during the plateaus even though heat is still being supplied.

Question 15 – A student investigates diffusion by placing one drop of red food colouring into two beakers containing equal amounts of still water.

- Beaker A contains cold water at 5°C.
- Beaker B contains warm water at 60°C.

The student observes that the red colour spreads throughout the warm water much faster than it spreads through the cold water.



Analyse the student's observation.

In your answer:

1. Explain why the food colouring spreads faster in the warm water using the kinetic theory of particles.
2. Predict what might happen if the experiment were repeated with the beaker placed on a vibrating surface, without changing the temperature.
3. Suggest one way the student could make the investigation a fair test.

IB MYP
Home Task (2026-27)
